

Wave Optics

Oscillations

Simple Harmonic Motion

$$\ddot{f}(t) = -\omega^2 f(t)$$

It is a constant coefficient 2nd-order ordinary differential equation

- It can be solved by an exponential function
Where ω is called angular frequency, *rad/s*

Assume that the solution is an exponential solution:

$$f(t) = ke^{\mu t}$$

Where k and μ are two constants

If $f(t)$ is a solution, then $2f(t)$ must be a solution too

$\Rightarrow k$ is arbitrary

Plug into the equation

$$\dot{f} = k\mu e^{\mu t}, \ddot{f} = k\mu^2 e^{\mu t} = \mu^2 f = -\omega^2 f$$

we get $\mu^2 = -\omega^2$. Thus $\mu = \pm i\omega$

$\Rightarrow$ A general solution: $f_1(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$

Damped Oscillation

$$\ddot{f}(t) + 2\beta\dot{f}(t) + \omega_0^2 f(t) = 0$$

It is a constant coefficient 2nd-order ordinary differential equation

Solution:

$$f(t) = C_1 e^{-\beta t} e^{i\omega t} + C_2 e^{-\beta t} e^{-i\omega t}$$

Where the angular frequency is

$$\omega \equiv \sqrt{\omega_0^2 - \beta^2}$$

Under-damped: $\beta < \omega_0$

- The solution can be rewritten as $f(t) = Ae^{-\beta t} \cos(\omega t + \phi)$
- $Ae^{-\beta t}$ can be considered as a diminishing amplitude

Critical Damping: $\beta = \omega_0$

- $\omega = 0$, another linearly independent solution is needed.
- $f(t) = C_1 e^{-\beta t} + C_2 t e^{-\beta t}$

Over-damped: $\beta > \omega_0$

- No oscillation:

$$f(t) = C_1 e^{-(\beta-\mu)t} + C_2 e^{-(\beta+\mu)t}$$

- Where $\mu \equiv i\omega = \sqrt{\beta^2 - \omega_0^2}$ is a real parameter

Forced Oscillation

$$\ddot{f}(t) + 2\beta\dot{f}(t) + \omega_0^2 f(t) = D_0 \cos(\omega t)$$

It is an inhomogeneous linear differential equation

Solution:

$$f(t) = f_{homo}(t) + f_{sp}(t)$$

where $f_{homo}(t)$ is the solution of corresponding homogeneous equation:

$$f_{homo}(t) = C_1 e^{-\beta t} e^{i\omega t} + C_2 e^{-\beta t} e^{-i\omega t}$$

It is negligible in the long term because $f_{homo}(t)$ as $t \rightarrow \infty$

Special Solution

$f_{sp}(t)$ is one special solution that satisfies the whole equation:

$$\ddot{f}_{sp} + 2\beta\dot{f}_{sp}(t) + \omega_0^2 f_{sp}(t) = D_0 \cos(\omega t)$$

Guess that $f_{sp}(t) = a \cos(\omega t) + b \sin(\omega t)$ then solve for a and b .

Alternative form: $f_{sp}(t) = A \cos(\omega t + \phi)$

Where $A = \sqrt{a^2 + b^2} = \frac{D_0}{\sqrt{4\beta^2 \omega^2 + (\omega^2 - \omega_0^2)^2}}$ and $\tan \phi = -\frac{b}{a} = \frac{2\beta\omega}{\omega^2 - \omega_0^2}$

Resonance occurs at $\omega = \omega_r \equiv \sqrt{\omega_0^2 - 2\beta^2}$

For small damping, $\omega \approx \omega_0$

Fourier Transform

Not all oscillations are simple harmonic

Fundamental Principle: Every oscillation can be written as superposition of single-frequency (simple harmonic) oscillations

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(\omega) e^{i\omega t} d\omega$$

Fourier transform is the "weight" of each frequency component

$$\tilde{f}(\omega) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

For Fourier transform to exist, it requires that

$$\int_{-\infty}^{\infty} |f(t)|^2 dt < \infty$$

Wave Optics

Interference

S_1 and S_2 are two coherent wave sources

Constructive interference occurs when the path difference is an integral number of wavelengths

Destructive interference occurs when the path difference is a half-integral number of wavelengths

Consider two oscillations with a phase difference ϕ ,

$$E_1(t) = E_0 \cos(\omega t), E_2(t) = E_0 \cos(\omega t + \phi)$$

The resultant amplitude satisfies

$$\begin{aligned} E(t) &= E_1(t) + E_2(t) = E_0 [\cos(\omega t) + \cos(\omega t + \phi)] \\ &= 2E_0 \cos \frac{\omega t + \omega t + \phi}{2} \cos \frac{\omega t - (\omega t + \phi)}{2} \\ &= 2E_0 \cos \left[\omega t + \frac{\phi}{2} \right] \cos \frac{\phi}{2} \end{aligned}$$

Averaging in time of the amplitude square:

$$\langle E(t)^2 \rangle = 4E_0^2 \left\langle \cos^2 \left(\omega t + \frac{\phi}{2} \right) \right\rangle \cos^2 \frac{\phi}{2} = 2E_0^2 \cos^2 \frac{\phi}{2}$$

In terms of the incoming light,

$$\langle E_1^2(t) \rangle = \langle E_0^2 \cos^2 \omega t \rangle = E_0^2 \langle \cos^2 \omega t \rangle = \frac{1}{2} E_0^2$$

$$\langle E^2(t) \rangle = 4 \langle E_1^2(t) \rangle \cos^2 \frac{\phi}{2}$$

Thus, the intensity of double slit interference is

$$I = 4I_0 \cos^2 \frac{\phi}{2}$$

where $\phi = \frac{2\pi}{\lambda} d \sin \theta = kd \sin \theta$

Diffraction

Spherical Waves

Second common wave type

$$\frac{A}{r(\mathbf{x})} \cos[kr(\mathbf{x}) - \omega t + \phi]$$

Where: $r(\mathbf{x}) \equiv |\mathbf{x} - \mathbf{x}_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}$

Wavefronts: Spheres centered at source $\mathbf{x}_0$

Amplitude decay: $1/r$ from energy conservation

Energy conservation: Same energy spreads over larger sphere

Fraunhofer Diffraction

By Huygens' principle, each point in slit acts as source:

$$E(\mathbf{x}, t) = \int_{-\frac{a}{2}}^{\frac{a}{2}} dy_0 \frac{A}{r} \cos(kr - \omega t)$$

Far field approximations ($R \gg a$):

Amplitude factor:

$$\frac{1}{r} \approx \frac{1}{R}$$

Phase factor:

$$r \approx R - y_0 \sin \theta$$

I'm taking PC3247 yay so I will learn this

$$\begin{aligned} E(\mathbf{x}, t) &\approx \frac{A}{R} \int_{-\frac{a}{2}}^{\frac{a}{2}} dy_0 \cos(k(R - y_0 \sin \theta) - \omega t) \\ &= \frac{A}{R} \frac{1}{-k \sin \theta} \sin[kR - \omega t - ky_0 \sin \theta] \Big|_{-\frac{a}{2}}^{\frac{a}{2}} \\ &= \frac{A}{R} \frac{1}{-k \sin \theta} \left(-2 \cos(kR - \omega t) \sin \frac{ka \sin \theta}{2} \right) \\ &= \frac{A}{R} \frac{1}{k \sin \theta} 2 \cos(kR - \omega t) \sin \frac{ka \sin \theta}{2} \\ &= \frac{aA}{R} \cos(kR - \omega t) \operatorname{sinc} \frac{ka \sin \theta}{2} \end{aligned}$$

Diffraction Result

Final result:

$$E(\mathbf{x}, t) = \frac{aA}{R} \cos(kR - \omega t) \operatorname{sinc} \frac{ka \sin \theta}{2}$$

Where: $\operatorname{sinc}(\beta) \equiv \frac{\sin \beta}{\beta}$

Key feature: Amplitude depends on direction through $\operatorname{sinc} \frac{ka \sin \theta}{2}$

$$\Rightarrow I(\theta) \propto \operatorname{sinc}^2 \left(\frac{ka}{2} \sin \theta \right)$$

Polarisation

Electric field is a vector

Light is a transverse wave: $\mathbf{E}$ perpendicular to propagation direction

Polarisation: Direction of electric field oscillation

Define orthogonal unit vectors in transverse plane:

$\hat{e}_H$ (horizontal) and $\hat{e}_V$ (vertical)

Such that $\hat{e}_H \times \hat{e}_V = \hat{e}_k$

Linear Polarisation

Electric field oscillates along fixed direction:

$$\mathbf{E}(\mathbf{x}, t) = E_0 \cos(kz - \omega t + \phi)$$

Where: $\hat{e}_\theta \equiv \cos \theta \hat{e}_H + \sin \theta \hat{e}_V$

Field oscillates along direction θ

Note: $\theta' = \theta + \pi$ gives same polarisation

Most common type in everyday applications

Circular Polarisation

Electric field **rotates** around propagation axis

$$\mathbf{E}(\mathbf{x}, t) = \frac{E_0}{\sqrt{2}} [\cos(kz - \omega t + \phi) \hat{e}_H \pm \sin(kz - \omega t + \phi) \hat{e}_V]$$

Right-handed: - sign

(counterclockwise from receiver view)

Left-handed: + sign

(clockwise from receiver view)

Factor $1/\sqrt{2}$ ensures proper normalisation

Elliptical Polarisation

Most general form:

$$\mathbf{E}(\mathbf{x}, t) = \frac{E_0}{\sqrt{2}} [\cos \theta \cos(kz - \omega t + \phi_1) \hat{e}_H + \sin \theta \sin(kz - \omega t + \phi_2) \hat{e}_V]$$

Special cases:

$$(\theta, \phi_1, \phi_2) = \begin{cases} (\theta, \phi, \phi), & \text{Linear} \\ (\frac{\pi}{4}, \phi, \phi + \frac{\pi}{2}), & \text{Circular right} \\ (\frac{\pi}{4}, \phi, \phi - \frac{\pi}{2}), & \text{Circular left} \end{cases}$$

Amplitude & Intensity

Electric field of electromagnetic wave:

$$E(x, t) = E_0 \cos(kx - \omega t + \phi)$$

Amplitude E_0 , Units: V/m for electric field

Instantaneous intensity:

$$I(x, t) \equiv c\epsilon_0 |E(x, t)|^2 = c\epsilon_0 E_0^2 \cos^2(kx - \omega t + \phi)$$

Light Intensity:

$$I(x) \equiv \langle I(x, t) \rangle_t = \frac{1}{2} c\epsilon_0 E_0^2$$

where $\langle \rangle_t$ means averaging over time:

$$\langle \cos^2(\omega t) \rangle_t \equiv \frac{1}{T} \int_0^T dt \cos^2(\omega t) = \frac{1}{2\pi} \int_0^{2\pi} d\theta \cos^2 \theta = \frac{1}{2}$$

Plane Wave

A monochromatic plane wave is given by

$$A \cos(\mathbf{k} \cdot \mathbf{x} - \omega t + \phi)$$

Wave vector: $\mathbf{k} = k\hat{\mathbf{e}}_k$ carries magnitude and direction

Propagates in direction of $\mathbf{k}$

Dispersion relation: $\omega = c|\mathbf{k}|$

Wavefronts: Planes perpendicular to $\hat{\mathbf{e}}_k$

Gaussian Wave Packet

Gaussian wave packet:

$$\psi(x, t) = \frac{1}{(\pi a^2)^{1/4}} e^{-\frac{(x_0 - ct)^2}{2a^2}} e^{i(k_0 x - \omega_0 t)}$$

Its Fourier transform is also Gaussian:

$$\tilde{\psi}(k) = \frac{1}{(\pi \kappa^2)^{1/4}} e^{-\frac{(k - k_0)^2}{2\kappa^2}}$$

where $\kappa = \frac{1}{a}$

Demonstrates uncertainty principle:

Narrow in space (small a) $\leftrightarrow$ Broad in frequency (large κ)

Normalise Gaussians

Use the formula

$$\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$$

To find the normalization factor $A(a)$ of a Gaussian

$$G(x) = A(a)$$

Relativity

Special Relativity

Principle: All inertial frames are equivalent

Note: the transverse distances are the same in both inertial frames.
Observe that

$$-(c\Delta t)^2 + (\Delta x)^2 = -4D^2 = -(c\Delta t_0)^2$$

Or

$$-(c dt)^2 + (dx)^2 + (dy)^2 + (dz)^2 = -(c dt_0)^2 + (dx_0)^2 + (dy_0)^2 + (dz_0)^2$$
$$ds^2 = ds_0^2$$

An invariance between these two inertial frames

Lorentz Transformation

Lorentz boost by θ :

$$ct' = ct \cosh \theta - x \sinh \theta$$
$$x' = -ct \sinh \theta + x \cosh \theta$$

Lorentz invariance

$$ds'^2 = ds^2$$

Velocity of S' relative to S :

$$v = c \tanh \theta$$

Length contraction & Time dilation

The rate at which time flows depends on how fast the observer moves!

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} = \gamma t_0$$

Note: this is not just the subjective perception of time, but the objective measurable length of time!

Consider a rod whose length is L_* when measured in its own rest frame.

$$x' = \gamma(x - vt), t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

$$\Delta x' = \gamma(\Delta x - v\Delta t), \Delta t' = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right)$$

$$\Delta t' = 0, \Delta x' = L, \Delta t \neq 0, \Delta x = L_*$$

$$\begin{cases} L &= \gamma(L_* - v\Delta t) \\ 0 &= \gamma(\Delta t - \frac{v}{c^2}L_*) \Rightarrow \Delta t = \frac{v}{c^2}L_* \end{cases}$$

$$L = \frac{L_*}{\gamma}$$

Twin Paradox

Astronaut Alice takes a space trip at a speed of $\beta = 0.65$ to a space station located 3.25 light-years from Earth and back starting on the 20th birthday while her twin brother Bob stays at home on Earth. The round trip distance is 6.5 light-years, so from Bob's point of view on Earth, Alice's trip takes

$$\frac{6.50}{0.65} = 10 \text{ years}$$

So when Alice returns, Bob is $20 + 10 = 30$ years old

Length Contraction

The gamma parameter for Alice is

$$\gamma = \frac{1}{\sqrt{1 - 0.65^2}} = 1.315$$

From Alice's point of view, she travels a length contracted distance

$$\frac{6.5}{1.315} = 4.942$$

From Alice's point of view, the trip takes

$$\frac{4.942}{0.65} = 7.6 \text{ years}$$

When she returns, Alice is $20 + 7.6 = 27.6$ years old

The Paradox

Now let's look at Bob from Alice's point of view

He seems to be sitting still and Bob is moving with speed $\beta = 0.65$

Alice ages 7.6 years during her trip, but Bob was moving with $\beta = 0.65$ ($\gamma \approx 1.316$) so his clock was running more slowly and he should have aged

$$\frac{7.6}{1.316} = 5.8 \text{ years}$$

So when Alice returned to Earth, she would be 27.6 years old and Bob would be 25.8 years old

Both twins can't be younger than the other!
Which of these two views is right?

Resolution of the Twin Paradox

The resolution of the twin paradox comes when we realize that Bob remains in an inertial frame on Earth while Alice lives in two different inertial frames during her trip

An outbound leg where she moves away from Earth and towards the space station

An inbound leg where she moves back from the space station towards Earth

The symmetry between the twins is broken!

We can analyze the path of the two twins by counting "Happy Birthday" signals they received from each other

Addition of velocities

Lorentz transformation

$$\begin{cases} t = \gamma \left(t' - \frac{vx'}{c^2} \right) \\ x = \gamma(x' - vt') \end{cases}$$

Differentiating, we get

$$\begin{cases} dt = \gamma \left(dt' - \frac{vdx'}{c^2} \right) \\ dx = \gamma(dx' - vdt') \end{cases}$$

taking $\frac{dx}{dt}$, we get

$$\frac{dx}{dt} = \frac{\left(dt' - \frac{vdx'}{c^2} \right)}{dx' - vdt'} = \frac{\left(1 - \frac{vdx'}{dt'c^2} \right)}{\frac{dx'}{dt'} - v}$$

Rearranging for $\frac{dx'}{dt'}$ and using $v_{1+2} = \frac{dx'}{dt'}$ and $u = \frac{dx}{dt}$, we get

$$v_{1+2} = \frac{u + v}{1 + \frac{uv}{c^2}}$$

Energy-momentum relation

Momentum and Energy (1)

Start with the results for momentum and energy and square:

$$\vec{p} = \gamma m \vec{v} \Rightarrow p^2 = \gamma^2 m^2 v^2$$

$$E = \gamma mc^2 \Rightarrow E^2 = \gamma^2 m^2 c^4$$

Square of the gamma-factor:

$$\gamma^2 = \frac{1}{1 - \beta^2} = \frac{1}{1 - \frac{v^2}{c^2}} = \frac{c^2}{c^2 - v^2}$$

Express E^2 as a function of p^2 :

$$E^2 = \frac{c^2}{c^2 - v^2} m^2 c^4 = m^2 c^4 + \frac{v^2}{c^2 - v^2} m^2 c^4$$

$$m^2 c^4 + \frac{c^2}{c^2 - v^2} m^2 v^2 c^2 = m^2 c^4 + \gamma^2 m^2 v^2 c^2$$

$$E^2 = m^2 c^4 + p^2 c^2$$

Momentum and Energy (2)

Finally we take the square root

$$E = \sqrt{p^2 c^2 + m^2 c^4}$$

Special case with no motion:

$$E = mc^2$$

Negative root: antimatter (Dirac)

$$E = -\sqrt{p^2 c^2 + m^2 c^4}$$

Case of zero mass (photons):

$$E = pc$$

Speed, Energy, and Momentum

Divide momentum by energy

$$\frac{p}{E} = \frac{\gamma m v}{\gamma m c^2} = \frac{v}{c^2} \Rightarrow \frac{pc^2}{E}$$

This gives us an additional energy-momentum relationship:

$$\beta = \frac{pc}{E} \Rightarrow p = \frac{\beta E}{c} \text{ or } E = \frac{pc}{\beta}$$

Doppler effect & Relativistic beaming

Lorentz boost:

$$\omega' = \gamma(\omega + vk^x), \quad k^{x'} = \gamma\left(k^x + \frac{v\omega}{c^2}\right)$$

$$|\vec{k}| = \frac{\omega}{c}, \quad k^x = |\vec{k}| \cos \alpha$$

with α the angle with the x axis

$$\Rightarrow \begin{cases} \omega' = \gamma \left(\omega + \frac{v\omega}{c} \cos \alpha \right) = \gamma \left(1 + \frac{v}{c} \cos \alpha \right) \omega \dots (1) \\ \frac{\omega'}{c} \cos \alpha' = \gamma \left(\frac{\omega}{c} \cos \alpha + \frac{v\omega}{c^2} \right) \rightarrow \omega' \cos \alpha' = \gamma \left(\cos \alpha + \frac{v}{c} \right) \omega \dots (2) \end{cases}$$

Cancel $\cos \alpha$:

$$(1) - \frac{v}{c}(2) \Rightarrow \omega' - \frac{v}{c}\omega' \cos \alpha' = \gamma \omega \left(1 - \frac{v^2}{c^2} \right) = \omega \sqrt{1 - \frac{v^2}{c^2}}$$

Doppler shift:

$$\Rightarrow \omega' = \frac{\sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c} \cos \alpha'} \omega$$

Relativistic Beaming

A uniformly radiating body moving toward you is brighter than if it is moving away

$$\frac{2}{1} \Rightarrow \cos \alpha' = \frac{\frac{v}{c} + \cos \alpha}{1 + \frac{v}{c} \cos \alpha}$$

Forward and Backward Emissions

Forward emission, $-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$,

$$\frac{v}{c} \leq \frac{v}{c} + \frac{\left(1 - \frac{v^2}{c^2}\right) \cos \alpha}{1 + \frac{v}{c} \cos \alpha} = \cos \alpha' = \frac{\cos \alpha + \frac{v}{c}}{1 + \frac{v}{c} \cos \alpha} = 1 - \frac{(c-v)(1 - \cos \alpha)}{c + v \cos \alpha} \leq 1$$

Backward emission, $-\pi \leq \alpha \leq -\frac{\pi}{2}$,

$$-1 \leq -1 + \frac{(c-v)(1 + \cos \alpha)}{c + v \cos \alpha}$$

General Relativity

Principle: All frames are equivalent

To simplify the analysis:

$$\frac{V}{c}, \frac{gh}{c^2} \text{ important}$$

$$\left(\frac{V}{c}\right)^2, \left(\frac{gh}{c^2}\right)^2 \text{ negligible}$$

Newtonian mechanics can be used!

Consider the emission of two successive light pulses by Alice and their reception by Bob

Suppose that $t=0$ is the time the first pulse is emitted, t_1 is the

time it is received, $\Delta\tau_A$ is the time the second pulse is emitted, and $t_1 + \Delta\tau_B$ is the time the second pulse is received

$$z_A(t) = h + \frac{1}{2}gt^2$$

$$z_B(t) = \frac{1}{2}gt^2$$

$$\Rightarrow \Delta\tau_B \approx \left(1 - \frac{gh}{c^2}\right)$$

Signal 1

$$A_1(t = 0, z = h) \rightarrow B_1\left(t_1, \frac{1}{2}gt_1^2\right)$$

Signal 2

$$A_2\left(\Delta\tau_A, h + \frac{1}{2}g\Delta\tau_A^2\right) \rightarrow B_2\left(t_1 + \Delta\tau_B, \frac{1}{2}g(t_1 + \Delta\tau_B)^2\right)$$

Signal 1

$$\begin{aligned} ct_1 &= h - \frac{1}{2}gt_1^2 \Rightarrow \frac{1}{2}gt_1^2 + ct_1 - h = 0 \\ \Rightarrow t_1 &= \frac{-c \pm \sqrt{c^2 + 2gh}}{g} \Rightarrow t_1 = \frac{c}{g} \left[\sqrt{1 + \frac{2gh}{c^2}} - 1 \right] \end{aligned}$$

Signal 2

$$\begin{aligned} c(t_1 + \Delta\tau_B - \Delta\tau_A) &= h + \frac{1}{2}g\Delta\tau_A^2 - \frac{1}{2}g(t_1 + \Delta\tau_B)^2 \\ \Rightarrow \frac{1}{2}g(t_1 + \Delta\tau_B)^2 + c(t_1 + \Delta\tau_B) - \left(h + \frac{1}{2}g\Delta\tau_A^2 + c\Delta\tau_A\right) &= 0 \\ t_1 + \Delta\tau_B &= \frac{-c \pm \sqrt{c^2 + 2g\left(h + \frac{1}{2}g\Delta\tau_A^2 + c\Delta\tau_A\right)}}{g} \\ \Delta\tau_B = t_1 + \Delta\tau_B - t_1 &= \frac{c}{g} \left[\sqrt{1 + \frac{2gh}{c^2} \left(1 + \frac{c\Delta\tau_A}{h} + \frac{1}{2} \frac{g\Delta\tau_A^2}{h}\right)} - \sqrt{1 + \frac{2gh}{c^2}} \right] \\ \sqrt{1+x} &\approx 1 + \frac{1}{2}x - \frac{1}{8}x^2 \\ &\approx \frac{c}{g} \left[1 + \frac{gh}{c^2} \left(1 + \frac{c}{h}\Delta\tau_A + \frac{1}{2} \frac{g\Delta\tau_A^2}{h}\right) - \frac{1}{8} \left(\frac{2gh}{c^2}\right)^2 \left(1 + \frac{c\Delta\tau_A}{h} + \dots\right)^2 - \left(1 + \frac{gh}{c} - \frac{1}{8} \left(\frac{2gh}{c^2}\right)^2\right) \right] \\ &= \frac{c}{g} \left[\frac{g\Delta\tau_A}{c} + \frac{1}{2} \frac{g^2\Delta\tau_A^2}{c^2} + \frac{1}{2} \frac{g^2h^2}{c^4} - \frac{1}{2} \frac{g^2h^2}{c^4} \left(1 + \frac{c^2\Delta\tau_A^2}{h^2} + \frac{1}{4} \frac{g^2\Delta\tau_A^4}{h^2} + \frac{2c\Delta\tau_A}{h} + \frac{cg\Delta\tau_A^3}{h^3}\right) \right] \end{aligned}$$

$$\begin{aligned}
\Delta\tau_B &\approx \frac{c}{g} \left[\frac{g\Delta\tau_A}{c} - \frac{g^2h^2}{2c^4} \left(\frac{g^2\Delta\tau_A^4}{4h^2} + 2\frac{c\Delta\tau_A}{h} + \frac{gc\Delta\tau_A^3}{h^3} \right) \right] \\
&= \Delta\tau_A - \frac{1}{8} \frac{g^3}{c^3} \Delta\tau_A^4 - \frac{gh}{c^2} \Delta\tau_A - \frac{g^2h}{2c^3} \Delta\tau_A^2 - \frac{1}{2} \frac{g^2}{c^2h} \Delta\tau_A^4 \\
&= \Delta\tau_A \left[1 - \frac{1}{8} \left(\frac{g\Delta\tau_A}{c} \right)^3 - \frac{gh}{c^2} - \frac{1}{2} \left(\frac{g\Delta\tau_A}{c} \right)^2 \right] \\
&\approx \Delta\tau_A \left(1 - \frac{gh}{c^2} \right)
\end{aligned}$$

gh : Gravitational Potential

Gravitational redshift

The crests of a light wave of definite frequency can be thought of as a series of signals emitted at a rate that is the frequency of the wave.

$$\omega_\infty = \left(1 - \frac{GM}{Rc^2} \right) \omega_\star$$

Twin paradox tested (1971)

$$\Delta\tau = \frac{1}{c^2} \int dt \{ gh(t) - \frac{1}{2} V_g(t) [V_g(t) + 2V_\otimes] \}$$

Black holes

Schwardzschild Radius

$$R_s = \frac{2GM}{c^2}$$

What happens if the gravitational lensing is so strong that light cannot escape the lens.

Quantum Mechanics

Light as Particles

Blackbody radiation

Photoelectric effect

Light may kick electrons out of metals.

Higher intensity, more electrons.

However, blue light striking cesium causes the cesium to emit electrons. Red light does not, regardless of light intensity.

Light comes in photons.

To emit an electron, the cesium atom must absorb a single photon whose energy exceeds the ionization energy of the outermost electron in cesium.

A blue photon has enough energy; a red photon does not.

Each photon carries energy:

$$E = hf = \hbar\omega$$

Energy-momentum conservation

Compton Scattering

Photon carries momentum too.

$$\mathbf{p} = \hbar\mathbf{k}$$

Momentum conservation

$$\mathbf{p} = \mathbf{p}' + \mathbf{P}_e$$

Energy conservation

$$h\frac{c}{\lambda} + m_e c^2 = h\frac{c}{\lambda'} + E_e$$

where $E_e^2 = m_e^2 c^4 + P_e^2 c^2$

Wavelength shift

$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos\phi)$$

Principles of QM

Wavefunctions

Wavefunctions contain all quantum information about a system's state and evolution

Wavefunctions can be complex-valued and with multiple components

Governed by the Schrodinger equation in non-relativistic quantum mechanics

Not directly measurable - serve as mathematical framework for calculating observable quantities

Upon measurement, wavefunctions collapse to definite states

Measurements yield discrete, quantized values

The system remains in the measured state post-collapse

Wave-particle duality: while the wavefunction evolves continuously, observations produce discrete outcomes.

Born Rule

Wavefunctions are probabilistic amplitude: at time t , the probability of finding the particle in the interval $(x, x + dx)$ is

$$dP(x, t) = |\Psi(x, t)|^2 dx = \Psi^*(x, t) \Psi(x, t) dx$$

This is the Born Rule

Normalisation/Unitarity:

$$1 = \int dP(x, t) = \int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx$$

Unitary implies particle number conservation

Bra-ket Notation

Define ket states as column vectors:

$$|H\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |V\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Then a θ -direction ket state is

$$|\theta\rangle = \cos \theta |H\rangle + \sin \theta |V\rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

A bra state is defined as the Hermitian (Dirac) conjugate of its corresponding ket state:

$$|\alpha\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \Rightarrow \langle \alpha| \equiv (|\alpha\rangle)^\dagger \equiv \begin{pmatrix} \alpha_1^* \\ \alpha_2^* \end{pmatrix}$$

Hermitian conjugate = Transpose + complex conjugate

Polarisation States

Inner Product

The inner product between a pair of bra- and ket-states is calculated using matrix multiplication rules:

$$\langle \alpha || \beta \rangle \equiv (|\alpha\rangle)^\dagger |\beta\rangle = \alpha_1^* \beta_1 + \alpha_2^* \beta_2$$

It is a complex number in general

Born rule: The above product of normalised states is a probability amplitude

Namely, its absolute square is the probability:

$$P(\text{finding } \alpha \text{ given } \beta) = |\langle \alpha || \beta \rangle|^2$$

Orthonormal Basis

Basis vectors are orthonormal (orthogonal and normalised):

$$\langle H|H\rangle = 1 = \langle V|V\rangle, \langle H|V\rangle = 0 = \langle V|H\rangle \langle V|H\rangle$$

they form a complete basis set

That is, any state vector can be expressed as a linear combination of the basis vectors:

$$|\alpha\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \alpha_1|H\rangle + \alpha_2|V\rangle$$

Polarised state $|\theta\rangle$ is normalised:

$$\langle\theta|\theta\rangle \equiv (|\theta\rangle)^\dagger|\theta\rangle = \cos^2\theta + \sin^2\theta = 1$$

Classical vs. Quantum Description of Polarisation

Classical picture: When a light beam passes through a θ -oriented polariser, it emerges as θ -polarised light

Applying a subsequent horizontal polariser reduces the intensity by a factor of $\cos^2\theta$

Quantum picture: For an initially θ -polarised photon, the probability of detection after passing through a horizontal polariser is

$$P(\text{finding H given } \theta) = |\langle H|\theta\rangle|^2 = \cos^2\theta$$

Same factor!

Change of Basis

Any linearly independent (and complete) set of vectors form a basis. Among all possible basis choices, orthonormal sets prove most practical and mathematically convenient

For example, the θ -direction polarisation and its perpendicular direction form an orthonormal basis:

$$|\theta\rangle = \cos\theta|H\rangle + \sin\theta|V\rangle$$

$$|\theta_\perp\rangle \equiv |\theta + \frac{\pi}{2}\rangle = -\sin\theta|H\rangle + \cos\theta|V\rangle$$

They are orthonormal

$$\langle\theta|\theta\rangle = 1 = \langle\theta_\perp|\theta_\perp\rangle, \langle\theta|\theta_\perp\rangle = 0$$

Inverse Transformation

Conversely, each vector in the old basis can be written as a linear combination of the new basis vectors:

$$|\theta\rangle = \cos\theta|H\rangle + \sin\theta|V\rangle$$

$$|\theta_\perp\rangle = -\sin\theta|H\rangle + \cos\theta|V\rangle$$

$$|H\rangle = \cos\theta|\theta\rangle - \sin\theta|\theta_\perp\rangle$$

$$|V\rangle = \sin \theta |\theta\rangle + \cos \theta |\theta_{\perp}\rangle$$

Completeness

For an arbitrary state $|\alpha\rangle = \cos \alpha |H\rangle + \sin \alpha |V\rangle$, we have

$$\begin{aligned} |\alpha\rangle &= (\cos \alpha \cos \theta + \sin \alpha \sin \theta) |\theta\rangle + (-\cos \alpha \sin \theta + \sin \alpha \cos \theta) |\theta_{\perp}\rangle \\ &= \cos(\alpha - \theta) |\theta\rangle + \sin(\alpha - \theta) |\theta_{\perp}\rangle \end{aligned}$$

Transformation of Quantum States

Wigner's Theorem: All transformations of quantum states must be linear and unitary

Linearity: The transformation preserves linear combinations

Unitarity: Inner products remain invariant under transformation

Matrix Representation: Due to linearity, any transformation can be represented by a square matrix

Unitary condition: This matrix U is unitary: $U^{\dagger}U = I$ where I is the identity matrix

Linear algebra in two dimension

Puzzles of QM

Schrodinger's cat

"Cat state"

$$\Psi_{cat} = \frac{1}{\sqrt{2}}(|alive\rangle + |dead\rangle)$$

It's neither alive or dead

Recall polarisation of a photon:

$$\Psi_{photon} = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$$

Can you say its $|H\rangle$ or $|V\rangle$ or both?

Double-slit in QM

Waves vs Particles in Classical Physics

Waves -> interference

- Two coherent sources -> interference fringes
 - Constructive interference: peaks align -> brighter
 - Destructive interference: peaks in opposite phase -> cancels
- Particles -> no interference

- Particles go through slit A or slit B
- Expected pattern on screen: two broad humps
- No alternating bright/dark fringes

Double-Slit with Single Quanta

Experiment setup

- Source emits one electron/photon at a time
 - Barrier with two narrow slits
 - Distant screen detects impacts
- Expectation from classical physics: Two humps
- Observations
- Each detection is a single dot
 - Dots accumulate into a pattern
 - With both slits open: interference fringes

Classical particles

Probabilities add up in classical physics

With only one slit open, we observe a single peak

With both slits open, we observe a straightforward sum of two peaks

Quantum Particles

Each slit has a diffraction effect

Two slits have interference

$\text{Prob}_{A+B} \neq \text{Prob}_A + \text{Prob}_B$

Rather, $\Psi_{A+B} = \Psi_A + \Psi_B$

Path Detectors

Place detectors at slits to record "A" or "B"

Now each particle yields:

- a dot on screen
- plus a which-path record

Result:

- Interference fringes disappear
- Pattern becomes $\text{Prob}_A + \text{Prob}_B$

Explanation

- Interference needs coherence
- Observation destroy coherence

Mach-Zehnder interferometer

Beam Splitter

A beam splitter is a semi-transparent mirror.

It reflects half of the incident light and transmits the other half.

In bra-ket notation, a perfect beam splitter performs the transformation:

$$|1\rangle \rightarrow \frac{1}{\sqrt{2}}(|3\rangle + i|4\rangle), |2\rangle \rightarrow \frac{1}{\sqrt{2}}(i|3\rangle + |4\rangle)$$

The imaginary coefficients above are essential to ensure the transformation remains unitary, preserving probability conservation

Unitary Transformation

Re-labeling all horizontally propagating beams as $|x\rangle$ and vertically propagating beams as $|y\rangle$.

The previous transformation becomes

$$|x\rangle \rightarrow \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle), |y\rangle \rightarrow \frac{1}{\sqrt{2}}(i|x\rangle + |y\rangle)$$

Matrix representation:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$

Clearly, it won't be unitary if all matrix elements are equal

Mach-Zehnder Interferometer

A Mach-Zehnder Interferometer consists of two beam splitters, two mirrors, and a transparent phase shifter placed in one arm

The first beam splitter splits the beam into two paths

The mirrors redirect the beams toward the second beam splitter, which recombines them

The phase shifter introduces an additional phase delay in one path

The second beam splitter can be removed to obtain which-path info

The outputs are measured by two detectors: D_1 for the x -port, and D_2 for the y -port

Math Modelling

Initial state: $|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Beam splitter:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$

Mirror: swaps the x and y modes

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Phase shifter: adds phase only to x component:

$$F = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & 1 \end{pmatrix}$$

A Photon Goes through MZI

After the 1st BS, the photon state becomes

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Half in x path, half in y path (with a relative phase $\frac{\pi}{2}$)

Before the mirrors, the phase shifter acts as

$$\frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\phi} \\ i \end{pmatrix}$$

The mirrors then swap x and y components:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} i \\ e^{i\phi} \end{pmatrix}$$

Detecting Rates without the 2nd BS

Without the 2nd BS, D_1 detects the x-direction beam.

It will be triggered at the probability of

$$\left| \frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2} = 50\%$$

Similarly, D_2 is triggered at the chance of

$$\left| \frac{e^{i\phi}}{\sqrt{2}} \right|^2 = \frac{1}{2} = 50\%$$

Detecting Rates with the 2nd BS

If the 2nd BS is applied, the final state is

$$\frac{1}{2} \begin{pmatrix} i + ie^{i\phi} \\ -1 + e^{i\phi} \end{pmatrix}$$

D_1 detects at the rate of

$$\left| \frac{i}{2} (1 + e^{i\phi}) \right|^2 = \cos^2 \frac{\phi}{2}$$

Similarly, D_2 detects at the rate of $\sin^2 \frac{\phi}{2}$

Delocalisation

Varying the extra phase ϕ changes the detecting outcomes

- If $\phi = 0$, all photons exit in the x output
- If $\phi = \pi$, all photons exit in the y output

Changing ϕ in one arm changes the final state of every photon

The photon evolves as superposition of both paths

Delocalisation: The photon is delocalized across the interferometer

Delayed choice/quantum eraser: One may decide whether to insert the 2nd BS only after the photon has passed through the 1st BS - thereby choosing effectively, between observing interference (recombination) or retaining which-path information.

Entanglement

Some two-photon states can be rewritten as products of two single-photon states.

For example,

$$|HV\rangle = |H\rangle|V\rangle$$

$$a|HH\rangle + b|HV\rangle = |H\rangle(a|H\rangle + b|V\rangle)$$

However, some two-photon states cannot be factorised.

For example,

$$\frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle) \neq (a_1|H\rangle + b_1|V\rangle)(a_2|H\rangle + b_2|V\rangle)$$

A state that cannot be factorised in this way is called an entangled state.

EPR paradox

Einstein, Podolsky, and Rosen proposed a thought experiment challenging QM in 1935.

Here is a modified version.

Two photons are prepared in the entangled state $\frac{1}{\sqrt{2}}(|HV\rangle - |VH\rangle)$, and they travel in opposite directions

If Alice measures the polarisation of Photon 1 and obtains $|H\rangle$, then she immediately knows that Bob must obtain $|V\rangle$ if he measures Photon 2, and vice versa.

Does this imply information propagates faster than the speed of light?

No-cloning theorem

It is impossible to clone an unknown state without disturbing it. Suppose a physicist receives a photon in an unknown polarization state $|\psi\rangle$ and wants to prepare a second state with the same polarization.

To do so, he/she starts from a fixed "blank" state $|R\rangle$

One might hope to find a transformation such that

$$|R\rangle \rightarrow T(|R\rangle) = U|R\rangle = |\psi\rangle$$

Equivalently, cloning would require a process of the form

$$|\psi\rangle|R\rangle \rightarrow |\psi\rangle|\psi\rangle$$

Proof by Contradiction

Assume cloning is possible. Then for the basis states, we would have

$$|H\rangle|R\rangle \rightarrow |H\rangle|H\rangle$$

$$|V\rangle|R\rangle \rightarrow |V\rangle|V\rangle$$

Now, let's try to clone the superposition state $|+\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$

We start with $|+\rangle|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle|R\rangle + |V\rangle|R\rangle)$

By linearity, the cloning operation would give

$$|+\rangle|R\rangle \rightarrow \frac{1}{\sqrt{2}}(|H\rangle|H\rangle + |V\rangle|V\rangle) \neq |+\rangle|+\rangle$$

Hence, a universal cloning transformation cannot exist.

Quantum teleportation

Although an unknown state cannot be cloned, it can be teleported.

To do this, we use three photons.

Photon A is prepared by Alice in an arbitrary polarization state:

$$|\theta\rangle_A = \cos\theta|H\rangle_A + \sin\theta|V\rangle_A$$

Photons B & C are prepared in an entangled state and sent to Alice and Bob, respectively:

$$|\Phi^+\rangle_{BC} = \frac{1}{\sqrt{2}}(|H\rangle_B|H\rangle_C + |V\rangle_B|V\rangle_C)$$

Alice performs a joint (Bell-state) measurement on Photons A & B. This measurement projects A & B onto an entangled state and, in doing so, breaks the original entanglement between photons B & C.

Bell States

Bell States of two polarized photons:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle|H\rangle + |V\rangle|V\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|H\rangle|H\rangle - |V\rangle|V\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle|V\rangle + |V\rangle|H\rangle)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|H\rangle|V\rangle - |V\rangle|H\rangle)$$

They form an alternative (complete & orthonormal) basis for two photon states.

Bell-State Measurement

It can be shown that the three-photon state can be rewritten in the Bell basis states of A & B as

$$\begin{aligned} |\theta\rangle_A |\Phi^+\rangle_{BC} &= \frac{1}{2} |\Phi^+\rangle_{AB} (\cos\theta|H\rangle + \sin\theta|V\rangle)_C \\ &+ \frac{1}{2} |\Phi^-\rangle_{AB} (\cos\theta|H\rangle - \sin\theta|V\rangle)_C \\ &+ \frac{1}{2} |\Psi^+\rangle_{AB} (\sin\theta|H\rangle + \cos\theta|V\rangle)_C \\ &+ \frac{1}{2} |\Psi^-\rangle_{AB} (-\sin\theta|H\rangle + \cos\theta|V\rangle)_C \end{aligned}$$

Alice's Bell-state measurement on Photons A & B will collapse Photon C into the corresponding conditional state shown in (...)

Rearranging Photon States

$$\begin{aligned} |\theta\rangle_A |\Phi^+\rangle_{BC} &= (\cos\theta|H\rangle_A + \sin\theta|V\rangle_A) \frac{1}{\sqrt{2}} (|H\rangle_B|H\rangle_C + |V\rangle_B|V\rangle_C) \\ &= \frac{1}{\sqrt{2}} (\cos\theta|HHH\rangle + \cos\theta|HVV\rangle + \sin\theta|VHH\rangle + \sin\theta|VVV\rangle) \\ &= \frac{1}{\sqrt{2}} \left[\cos\theta \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{AB} + |\Phi^-\rangle_{AB}) |H\rangle_C + \sin\theta \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{AB} - |\Psi^-\rangle_{AB}) |V\rangle_C \right] \end{aligned}$$

Teleportation

Alice then sends Bob her Bell-state measurement results through a classical channel (only 2 bits info is needed).

Based on this result, Bob applies the corresponding unitary operation to Photon C.

After this correction, photon C is restored to exactly the same state that Photon A was in before the protocol.

Quantum teleportation does not violate special relativity, because completing the protocol requires classical communication, which cannot travel faster than light.

Unitary Transformation Needed by Bob

if Alice's result is Φ plus, yadda yadda yadda

A Brief History of Quantum Teleportation

Theoretical proposal: by Bennett, Brassard, Crepeau, Jozsa, Peres, & Wootters in 1993

First experimental demonstrations: independently by two groups, Popescu and Zeilinger in 1997

Platforms demonstrated: photons, atoms, electrons, superconducting circuits, etc

Record distance: Photon states have been teleported over 1,400 km using a satellite

Bennett & Brassard just won the Turing Award 2026 because of BB84 protocol for quantum key distribution

Bell inequality

Bell Theorem (1964)

Consider a setup similar to the EPR scenario: Two photons are prepared in an entangled state and sent in opposite directions to Alice and Bob.

Alice randomly chooses to measure Photon 1 is in either the $|\alpha\rangle$ basis or the $|\alpha'\rangle$ basis

She records the outcomes as $A_0 = \pm 1$ for positive/negative $|\alpha\rangle$, and $A_1 = \pm 1$ for positive/negative $|\alpha'\rangle$

Similarly, Bob randomly chooses to measure Photon 2 is in either $|\beta\rangle$ or $|\beta'\rangle$ polarisation. He records outcomes as $B_0 = \pm 1$ or $B_1 = \pm 1$

Bell theorem: if the photons carry fixed (predetermined) polarisations before measurements, then the correlations must satisfy the inequality:

$$|\langle A_0 B_0 \rangle - \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle + \langle A_1 B_1 \rangle| \leq 2$$

Proof of Bell Inequality

If the photons carry fixed (predetermined) polarizations, then Alice's outcomes are independent of Bob's measurement choice and outcome.

We can therefore analyse each run deterministically before taking

ensemble average.

Regroup the expression:

$$A_0B_0 - A_0B_1 + A_1B_0 + A_1B_1 = (A_0 + A_1)B_0 - (A_0 - A_1)B_1$$

Since both A_0 and A_1 can only take the values of ± 1 , we must have either $A_0 = A_1$ or $A_0 = -A_1$

Hence, one of the two parantheses, $(A_0 + A_1)$ or $(A_0 - A_1)$ must be zero

The remaining nonzero parantheses then equals $+ 2$ or $- 2$

Proof of Bell Inequality

Multiplying the non-zero parentheses by $B_0 = \pm 1$ or $B_1 = \pm 1$, the result is still ± 2

Averaging over many trials may lead to partial cancellation between positive and negative contributions, but the magnitude of the averaged result cannot exceed 2

Therefore, if photons have predetermined polarisations before measurement, the Bell inequality must hold:

$$|\langle A_0B_0 - A_0B_1 + A_1B_0 + A_1B_1 \rangle| = |\langle A_0B_0 \rangle - \langle A_0B_1 \rangle + \langle A_1B_0 \rangle + \langle A_1B_1 \rangle| \leq 2$$

However, we will see that QM violates Bell inequality

Bell Inequality in QM

Consider two photons in the Bell state, $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|H\rangle|V\rangle - |V\rangle|H\rangle)$

Re-express the 1st photon state in the basis of $|\alpha\rangle$ & $|\alpha^\perp\rangle$, and the 2nd photon in $|\beta\rangle$ & $|\beta^\perp\rangle$, we get

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}[(\cos \alpha |\alpha\rangle - \sin \alpha |\alpha_\perp\rangle)(\sin \beta |\beta\rangle + \cos \beta |\beta_\perp\rangle) - (\sin \alpha |\alpha\rangle + \cos \alpha |\alpha_\perp\rangle)(\cos \beta |\beta\rangle - \sin \beta |\beta_\perp\rangle)]$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(\sin \theta_{\alpha\beta} |\alpha\rangle |\beta\rangle + \cos \theta_{\alpha\beta} |\alpha\rangle |\beta_\perp\rangle - \cos \theta_{\alpha\beta} |\alpha_\perp\rangle |\beta_\perp\rangle + \sin \theta_{\alpha\beta} |\alpha_\perp\rangle |\beta\rangle)$$

where $\theta_{\alpha\beta} \equiv \beta - \alpha$

Probabilities

The probability amplitude of finding photon 1 in $|\alpha\rangle$ and photon 2 in $|\beta\rangle$ is

$$\langle \alpha | \langle \beta | \Psi^- \rangle = \frac{\sin \theta_{\alpha\beta}}{\sqrt{2}}$$

By Born rule, the probability is

$$\text{Prob}_{\alpha\beta} = |\langle \alpha | \langle \beta | \Psi^- \rangle|^2 = \frac{1}{2} \sin^2 \theta_{\alpha\beta}$$

Similarly,

$$\text{Prob}_{\alpha\beta_{\perp}} = |\langle\alpha|\langle\beta_{\perp}|\Psi^{-}\rangle|^2 = \frac{1}{2}\cos^2\theta_{\alpha\beta}$$

$$\text{Prob}_{\alpha_{\perp}\beta} = |\langle\alpha_{\perp}|\langle\beta|\Psi^{-}\rangle|^2 = \frac{1}{2}\cos^2\theta_{\alpha\beta}$$

$$\text{Prob}_{\alpha_{\perp}\beta_{\perp}} = |\langle\alpha_{\perp}|\langle\beta_{\perp}|\Psi^{-}\rangle|^2 = \frac{1}{2}\sin^2\theta_{\alpha\beta}$$

Expectation Values

Therefore, the expectation value of A_0B_0 is

$$\langle A_0B_0 \rangle = -\cos(2\theta_{\alpha\beta})$$

Bell-CHSH Inequality

Similarly,

$$\langle A_0B_1 \rangle = -\cos(2\theta_{\alpha\beta}), \langle A_1B_0 \rangle = -\cos(2\theta_{\alpha'\beta}), \langle A_1B_1 \rangle = -\cos(2\theta_{\alpha'\beta'})$$

Plugging in the LHS of Bell-CHSH inequality, we obtain,

$$S \equiv \langle A_0B_0 \rangle - \langle A_0B_1 \rangle + \langle A_1B_0 \rangle + \langle A_1B_1 \rangle = -\cos(2\theta_{\alpha\beta}) + \cos(2\theta_{\alpha\beta'}) - \cos(2\theta_{\alpha'\beta}) - \cos(2\theta_{\alpha'\beta'})$$

Note that not all θ are independent:

$$\theta_{\alpha\beta'} = \beta' - \alpha = (\beta' - \alpha') + (\alpha' - \beta) + (\beta - \alpha) = \theta_{\alpha'\beta'} - \theta_{\alpha'\beta} + \theta_{\alpha\beta}$$

Choose $\theta_{\alpha\beta} = -\theta_{\alpha'\beta} = \theta_{\alpha'\beta'} \equiv \theta$, we obtain $\theta_{\alpha\beta'} = 3\theta$ and a function

$$S(\theta) = -3\cos(2\theta) + \cos(6\theta)$$

QM Violates Bell Inequality

From the plot of $S(\theta)$, we can see there are regions that the Bell inequality is violated:

$$|S| > 2$$

The maximum violation occurs at $\theta = \frac{\pi}{8}, \dots$, giving

$$|S|_{max} = \left| -3\cos\frac{\pi}{4} + \cos\frac{3\pi}{4} \right| = 2\sqrt{2} \approx 2.828$$